A Blockchain-Integrated Framework for Secure, Incentivized, and Scalable Machine Learning: Integrating Consensus Mechanisms and Reinforcement Learning

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Abstract

The expansion of Machine Learning (ML) is driving innovation in healthcare, finance, and IoT, but traditional centralized approaches face security, privacy, and trust challenges. Federated Learning (FL) goes for a decentralized approach but is struggling with validation overhead, adversarial threats, and trust in model updates. Here, the study is integrating blockchain with FL and Reinforcement Learning to strengthen security and data integrity. The **Blockchain-Integrated Federated Learning (B-FL)** framework also leverages blockchain's immutable ledger, Byzantine Fault Tolerance (BFT), proof of Deep Learning (PoDL), and smart contracts. Moreover, privacy enhancing techniques such as Fragmented Federated Learning (FFL) and Additive Homomorphic Encryption (AHE) mitigating attacks such as model poisoning attacks and at the same time ensuring data aggregation. Nonetheless blockchain integration may introduce computational overhead, this study also optimizes consensus mechanisms and incentive structures to create a scalable, privacy- preserving ML framework applicable to healthcare, finance and IoT.

Problem Statement

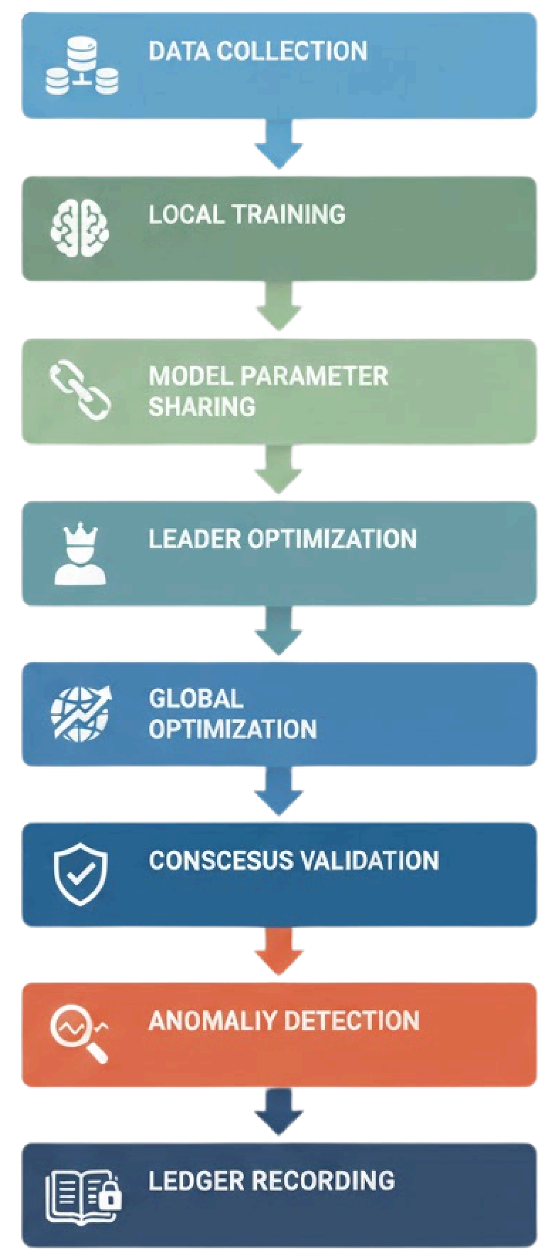
The rapid expansion of decentralized computing and financial digitization has intensified the demand for secure, privacy-preserving, and trustworthy machine learning frameworks. Traditional centralized ML architectures expose sensitive financial and transactional data to risks of leakage, manipulation, and regulatory non-compliance. Federated Learning (FL) mitigates these issues by enabling collaborative model training without direct data sharing; however, it still faces limitations in trust validation, aggregation integrity, and performance optimization. Blockchain technology introduces a verifiable, tamper-resistant infrastructure that can strengthen the transparency and accountability of FL. Yet, integrating blockchain and FL introduces new challenges, including high latency, inefficient consensus mechanisms, and limited adaptability in heterogeneous data environments. To address these gaps, this research proposes a Consortium Blockchain-Integrated Federated Learning Framework combining Hyperledger Fabric, Dynamic Butterfly-Billiards Optimization Algorithm (DB-BOA), and Adaptive Deep Temporal Context Network (ADTCN). The framework leverages blockchain's immutability and DB-BOA's meta-heuristic intelligence for optimized leader selection, consensus latency reduction, and hyperparameter tuning, while ADTCN provides adaptive temporal anomaly detection in financial transactions. Through secure smart contracts and privacy-preserving parameter exchange, the system aims to achieve high fraud-detection accuracy, low transaction latency, and resilience against adversarial or poisoning attacks. This study contributes to the advancement of secure, federated, and performance-optimized learning frameworks for consortium-based financial infrastructures.

Objective

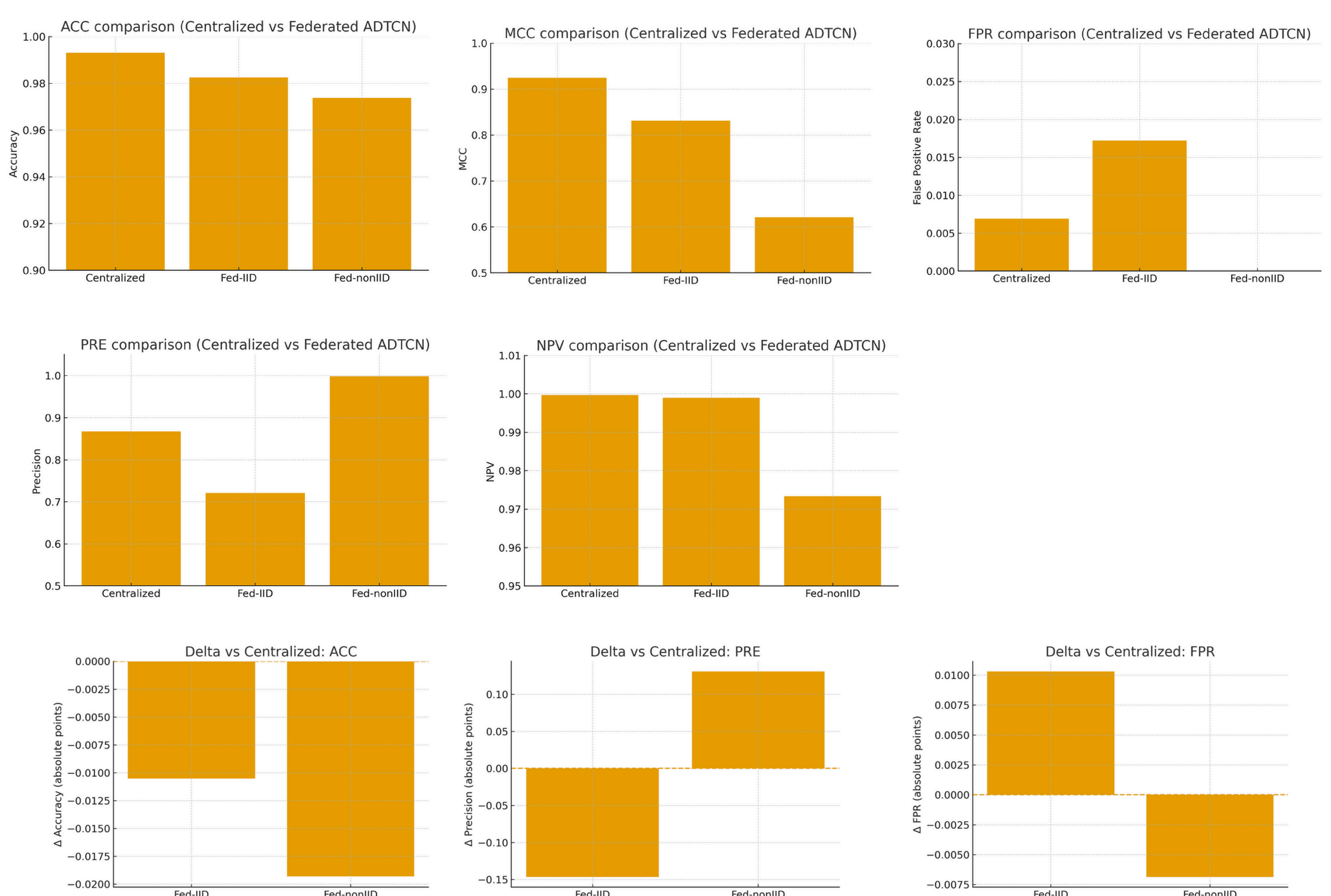
- Uses blockchain to secure data and manage model updates efficiently.
- Enhances trust and collaboration in Federated Learning.
- DB-BOA incorporates token incentives and reputation-based fitness scoring to discourage malicious activity and ensure fair, trustworthy leader and model selection across consortium nodes.
- **FL-ADTCN** for fraudulent activity detection
- Enables a decentralized, scalable ML system for healthcare, finance, and IoT.

Methodology Overview

- **Data Collection:** Financial transactions from consortium members (banks, gateways).
- **Local Training:** Each node trains ADTCN on its own data.
- **Model Sharing:** Trained weights sent securely to blockchain via PoA.
- **Leader Selection:** DB-BOA chooses the optimal leader for aggregation.
- **Global Optimization:** DB-BOA minimizes latency and tunes hyperparameters.
- **Consensus:** PoA validates and records verified updates only.
- **Anomaly Detection:** ADTCN detects temporal financial irregularities.
- **Ledger Storage:** Validated results stored immutably in Hyperledger Fabric's World State.



Graphical Analysis and Interpretations



Figures 1-6: Comparative performance of Centralized, Federated IID, and Federated non-IID ADTCN models. Centralized training consistently achieves the best results; Federated IID remains close with balanced accuracy and stability, while Federated non-IID shows precision dominance but reduced MCC due to client data heterogeneity.

Metric	Centralized	Federated (IID)	Federated (non-IID)
Accuracy	0.9931	0.9826	0.9738
MCC	0.9249	0.8314	0.6213
FPR	0.0069	0.0172	0.00004
NPV	0.9997	0.999	0.9734
Precision	0.8673	0.7207	0.998

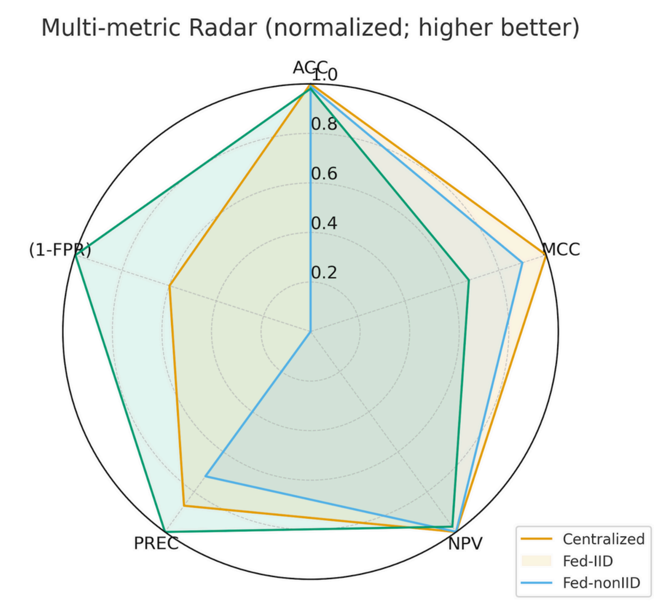


Figure 7: Multi-metric radar comparison of Centralized, Fed-IID, and Fed-nonIID modes. Centralized leads overall; Fed-IID stays balanced; Fed-nonIID shows high Precision but weaker MCC due to data imbalance.

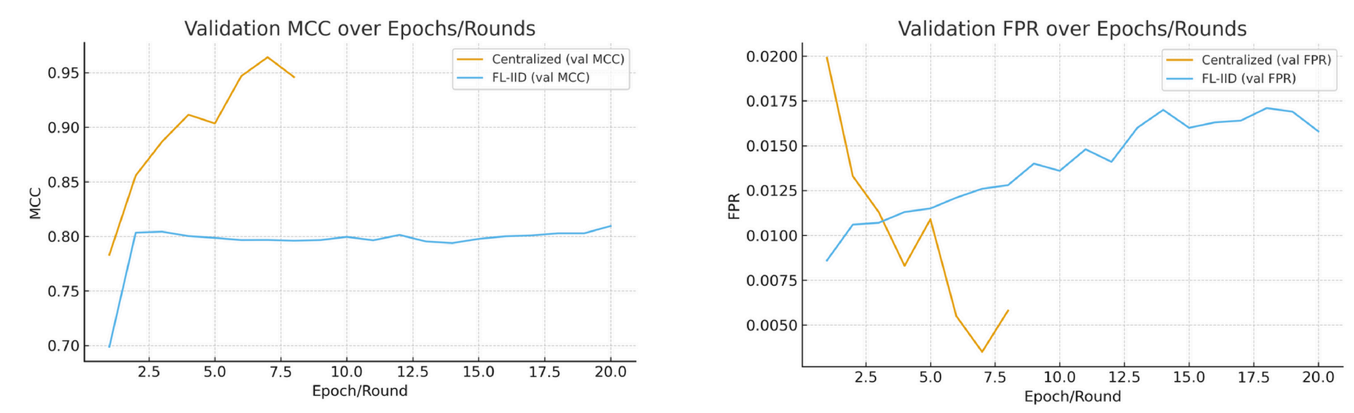


Figure 8: Validation curves comparing Centralized and Federated IID training. Centralized model converges faster with higher MCC and lower FPR, while Fed-IID stabilizes gradually—showing slower convergence but better consistency across rounds.

Constraints and Further Development

Type	Description	Impact	Planned Enhancement
Dataset	Synthetic/simplified transaction data used instead of live blockchain stream.	Limited realism.	Replace with real consortium financial datasets.
Computation	Single-node simulation of multi-agent optimization.	No true parallel search stress-test.	Move to distributed Fabric testbed.
Control Parameter	Switching probability PsP_sPs set empirically.	Potential sub-optimal switching rate.	Implement reinforcement-based adaptive control.
Incentive Layer	Reward mechanism for node performance not yet integrated.	Missing economic feedback loop.	Introduce blockchain-embedded incentive model.

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